DAY 1 · DATA PREPARATION

Smart Room Data Preprocessing

Room occupancy detection from temperature, humidity, light and CO2 | 11 October 2026

SCOPE

Identify processing needs from the inspection

- Recommend treatment of missing values, duplicates and outliers
- Prepare the date column
- Propose a chronological split that avoids data leakage

1 Processing needs and treatment

Inspection finding	Recommended action
No missing values in existing rows	No imputation is needed. Do not create values or occupancy labels for unobserved periods.
No exact duplicate rows or timestamps	No duplicate removal is needed. Preserve repeated sensor values recorded at different times.
Unusual sensor readings	Retain unusual readings for now. Remove or correct them only when there is evidence of a measurement error.
Numerical sensor inputs	Scale numerical inputs for logistic regression. Random forest does not require scaling.

2 Date preparation and gaps

- The date column was converted from object to datetime64[ns] and sorted.
- The recording period is 2 February 2015 at 14:19 to 18 February 2015 at 09:19.
- Ordinary intervals are 59, 60 or 61 seconds, approximately one reading per minute.
- Two gaps occur. Their cause is unknown.

Gap	Last reading before	First reading after	Duration
Gap 1	4 February, 10:43	4 February, 17:51	7 h 8 min
Gap 2	10 February, 09:33	11 February, 14:48	29 h 15 min

Gap handling. Leave recording gaps unfilled because sensor readings and occupancy during these periods are unknown. Do not count the missing periods as occupied or empty.

Weekday schedule. The room was recorded as unoccupied throughout both weekends, 7–8 and 14–15 February 2015. This suggests that room usage may follow a weekday schedule.

3 Proposed chronological split

The data is split by time, in order, with no shuffling. Each set comes after the one before it.

Train 2–11 Feb	Validation 12–15 Feb	Test 16–18 Feb

Dataset	Period in 2015	Rows
Train	2–11 February	11,360
Validation	12–15 February	5,760
Test	16–18 February	3,440

All dates are in 2015. The split uses chronological row order; all subsets contain both classes.

4 Data leakage controls

1	Fit scaling and any learned imputation or clipping on Train only.
2	Use Validation for model and setting choices.
3	Use Test for final evaluation.
4	Exclude Occupancy from X.

5 Feature preparation and scaling

Step 1 Separate inputs and target

After the chronological split, separate the five sensor inputs (X) from Occupancy (y) in Train, Validation and Test. Keep date for ordering and analysis; do not scale the date or the target.

Step 2 Scale with StandardScaler

Use StandardScaler for Temperature, Humidity, Light, CO2 and HumidityRatio. Fit it on X_train only, then use the same fitted scaler to transform X_val and X_test. Each feature is centered using its training mean and scaled using its training standard deviation.

Step 3 Choose the features for each model

Model	Input features
Logistic regression	Scaled features (X_train_scaled, X_val_scaled, X_test_scaled)
Random forest	Original sensor features (X_train, X_val, X_test)

Negative scaled values are normal and indicate readings below the training mean.

Code

```
from sklearn.preprocessing import StandardScaler

features = ["Temperature", "Humidity", "Light",
            "CO2", "HumidityRatio"]

X_train, y_train = train[features], train["Occupancy"]
X_val, y_val = validation[features], validation["Occupancy"]
X_test, y_test = test[features], test["Occupancy"]

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_val_scaled = scaler.transform(X_val)
X_test_scaled = scaler.transform(X_test)
```